Markus Gruber, M.Sc.

Senior C++/Python Backend Systems Engineer

High-Performance Architectures · **Pragmatic Problem Solver** · **Domain Expert**

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Professional Profile



Pragmatic Senior C++/Python Backend & Systems Engineer with 9+ years of experience building real-time pipelines, cloud microservices, and high-throughput systems. Holds an M.Sc. in Scientific Computing with a strong mathematical foundation.

Resilient problem solver resolving critical production bottlenecks and memory leaks. Acts as an effective interface between development, QA, and architecture while leveraging modern AI tooling (Cursor, Copilot).

Engineering Profile

TECHNICAL FOUNDATION & SYSTEM ARCHITECTURE DOMAINS

Backend Architecture	Systems & Profiling	Data & Verification
Modern C++ (11/14/17/20), Python	Memory Optimization & Profiling	PostgreSQL, DuckDB, Redis
REST APIs & Microservices	Linux Systems & Scripting	Automated Benchmarking Suites
Multithreading & Concurrency	Safety-Critical Systems (C)	CI/CD & Docker Pipelines

CORE TECHNICAL SPECIALIZATION – BACKEND & DISTRIBUTED SYSTEMS

- **High-Performance Backend Engineering:** Architecting low-latency microservices, REST APIs, and scalable C++/Python execution engines.
- **Distributed Data Pipelines & Systems:** Optimizing memory management, concurrent processing, and database interactions for production workloads.
- **Automated Verification & Telemetry:** Implementing end-to-end performance profiling, automated benchmarking suites, and telemetry frameworks.

Professional Experience

Mindbreeze GmbH – Software / Quality Engineer (C++/Python) 09/2022 – 07/2026

C++, Python, REST APIs, Linux, Docker, Azure DevOps, Telemetry & Performance Profiling | Team: 6 Engineers

- Engineered Python/Bash automation frameworks, reducing release verification duration from 10 working days down to 3 working days.
- Optimized performance-critical backend components for enterprise architectures handling distributed data processing.
- Refactored search engine core modules and optimized index data pipelines to improve high-throughput query execution.
- Primary technical troubleshooter during release phases, resolving memory leaks and performance degradations in live systems.

Untis GmbH – Software Engineer (C++) 02/2022 – 06/2022

C++, System Refactoring, Performance Profiling, Core Desktop Modules | Team: 10 Engineers

- Refactored legacy C++/MFC core modules using Visual Studio to eliminate technical debt, optimize memory allocation, and resolve production crashes.

logi.cals – Embedded Software Engineer (C/C++) 02/2021 – 01/2022
C, C++, Embedded Systems, Safety-Critical Architectures, Deterministic Execution | Team: 6 Engineers

- Engineered deterministic C/C++ runtime execution systems (Software PLCs) and automated verification test suites for safety-standard certification across hardware platforms.

EYYES – Software Engineer (C++/Python) 04/2017 – 12/2020
C++, Python, Real-Time Execution Pipelines, CUDA, Embedded GPU Edge | Team: 8 Engineers

- Developed computer vision and deep learning pipelines (Keras, ONNX) for industrial and embedded camera systems to ensure continuous real-time edge performance.
- Engineered low-latency C++ and CUDA execution routines for real-time video stream processing on embedded GPU edge devices.
- Implemented hardware-near software components in C++ to control camera sensors, stepper motors, and SerDes interfaces.
- Optimized detection accuracy and recognition precision by training multiple TensorFlow classifiers to 99.99% accuracy through rigorous hardware-near algorithm tuning and automated benchmarking suites.

Selected Software Projects

- **Magnum C++ Game of Life:** High-performance C++ simulation focusing on low-level memory layout, cache locality, and object-oriented architecture (GitHub).
  Tech: Modern C++, Memory Layout, Performance Optimization, Grid Logic
- **AWS Cloud Deep Learning Pipeline:** End-to-end cloud pipeline automating data ingestion and high-throughput PyTorch model inference (GitHub).
  Tech: AWS, Python, PyTorch, MLOps, REST APIs
- **Azure AI Document Analyzer & RAG Infra:** High-throughput document indexing and RAG retrieval backend deployed on Azure with vector databases (GitHub).
  Tech: Python, RAG, Vector DBs, Azure, Microservices

Technical Skills

Languages & Core: Modern C++, C, Python, SQL, Java, Bash, Low-Level Memory Management, Parallel Algorithms, Scientific Computing.
Architecture & Quality: Code Reviews, Technical Governance, Clean Code, CI/CD, Microservices, REST APIs, Performance Profiling.
Cloud, Systems & Infra: Amazon AWS (S3, RDS, EC2), Microsoft Azure, Linux (Ubuntu/Debian), Windows, Docker, Kubernetes/OpenShift Basics.
Data & ML Infrastructure: RAG Infrastructure, LangChain, PySpark, DuckDB, PostgreSQL, SQLAlchemy, CUDA, OpenCL.

Education

M.Sc. Scientific Computing – University of Vienna 10/2013 – 06/2017
Thesis: Silent Data Corruption in FT-GMRES (Link, Advisor: Prof. Dr. W. Gansterer).

B.Sc. Industrial Simulation – FH St. Pölten 09/2010 – 06/2013
Thesis: Simulation of Nanocomposite Materials (Link, Advisor: Prof. Dr. T. Schrefl).

Engineering Approach & Mindset

- **Systematic Problem Solving:** First-principles analytical approach to systematic debugging and stability optimization.
- **Quality & Efficiency Focus:** Pragmatic engineering mindset balancing rapid deployment with long-term maintainability and high performance.
- **Cross-Functional Integration:** Acts as the technical glue bridging development, QA, and architecture.

Languages & Work Status

German: Native (C2) · **English:** Full Professional Proficiency (C1/C2) · **Work Permit:** EU Citizen / Austria